A PROPOSAL FROM FINE ART MEDIA & RMIT UNIVERSITY

ARCHAI × ACMI

A sovereign, local-first AI collaboration — turning ACMI's collection into a *living archive* that visitors can talk to, in any language, while every record stays on hardware ACMI controls.

Prepared for the ACMI Registration & Collections and Technology teams 19 August 2026
Rob Graham · rob@fineartmedia.tech

01 The idea, in one breath

Cultural institutions have digitised millions of objects, yet most of that knowledge stays locked in cataloguing structures built for documentation, not discovery. Meanwhile the easy AI on offer is commercial cloud: powerful, but it takes an institution's collection data, visitor conversations and interpretive authority offshore.

ARCHAI is the alternative. It is an open-source semantic layer that sits *above* ACMI's existing collection and asset systems — it reads from them, never writes to them — and adds conversational interpretation, multilingual access, rights-aware search and staff-facing collection intelligence. It runs on institutionally owned hardware. The collection, and the new layer of visitor conversation it creates, stay in the building.

> THE RESEARCH
>
> **Cultivating a Living Archive: Designing Sovereign AI Infrastructure for Cultural Heritage** — a practice-based RMIT Design PhD (Rob Graham). A related paper was accepted for the 6th Summit on New Media Art Archiving at ISEA2026, Dubai. In practice-based research, building the working system *is* the research.

02 Technical specification

A semantic interface layer, not a new CMS. ARCHAI connects to whatever ACMI already runs via API, normalises the metadata, and makes the collection semantically searchable and conversationally accessible — no migration, nothing replaced.

Three-layer architecture

HERITAGE FOUNDATION

ACMI's own CMS / DAMS

Read-only. No AI output ever enters the record. Permanent, authoritative, citable.

DERIVATIVE PROCESSING

Vectors + AI responses

Entirely regenerable — rebuilt from unchanged source when models improve. Nothing precious lives here.

EDITORIAL

Curator interpretation

Curator-authored text flows one way, archive → visitor. It never writes back to the record.

The stack (all on-device)

SEMANTIC SEARCH

Qdrant + local embeddings

Rights- and place-aware; whole-collection queries for staff.

REASONING

Local LLM (Qwen)

Metadata-grounded; a five-layer framework keeps answers to verified fact.

VOICE

Whisper + MeloTTS

Speech in and out, on-device. A genuine Australian voice; multilingual.

ACCESS

Cloudflare Tunnel

CORS-locked HTTPS to the local box. No collection data leaves the building.

Six technical contributions

- **Vector embeddings for heritage discovery** — semantic search without depending on controlled vocabulary.
- **Provenance-tracked interpretation** — every generated answer records its derivation chain back to source.

- **Layered architecture** – permanent heritage held apart from regenerable AI processing.
- **Conversational objects (AUXIO)** – individual items speaking from their own validated metadata.
- **Five-layer hallucination prevention** – responses stay grounded under adversarial conditions.
- **Transferable hardware reference** – a self-deployable stack for the GLAM sector.

LIVE TODAY

~1,400 public AUXIO pages

Across institutions on five continents; 3,000+ staff-searchable records; every object open-licensed, rights-restricted media excluded at ingest.

BASE DEPLOYMENT

One-time capital

Owned outright, auditable, rebuildable from first principles – not a per-seat cloud subscription.

---

03 A pilot with ACMI

Scoped small and low-risk: a bounded set of ACMI-approved records – rights, terminology, cultural protocols and publication rules all set by your team. ARCHAI builds AUXIO conversational pages for that set and gives Collections staff semantic search across it. ACMI's open collection API makes the connection straightforward – and ARCHAI can build on your existing interpretation and hardware, including The Lens and its touch-points, rather than duplicating them. Three ACMI-record demos already run (the green zoetrope, a Reynaud praxinoscope strip, Muybridge's galloping horse).

> ETHICS & SAFETY, BUILT IN
>
> Public messages run through **Saftio**, an on-device safety layer: anyone in crisis is routed to real human help (never an AI), hate speech is held for curatorial review, and culturally sensitive material is held under your protocols. Verified visitor facts earn a ★ **green star** and are added to the object's record – community knowledge feeding the living archive. The curatorial team reviews and awards this in the ARCHAI app; the public page only displays it.

04 Research & collaboration pathways

This isn't a cold pitch. **RMIT is already ACMI's Major University Partner** — a decade-long relationship, renewed in 2026, spanning student work-integrated learning, exhibitions, and an existing RMIT × ACMI ARC Fellowship. ARCHAI extends that partnership into sovereign AI research, with several ways to engage:

- **PhD research site & case study** — ACMI as the institutional partner for Rob's practice-based RMIT PhD; born-digital and screen-culture collections are where ARCHAI's preservation and interpretation work matters most.
- **ARC Linkage Project** — RMIT + ACMI as partners (A$50k–300k/yr over 2–5 years). ACMI's Major-Partner status materially strengthens a Linkage bid.
- **Visitor studies & evaluation** — ACMI as a site for evaluating conversational access and the living-archive model, feeding the Research-through-Design cycle.
- **Cultural protocols & Indigenous data governance** — co-designing the cultural-sensitivity layer with ACMI's First Nations protocols and ICIP, as enforceable characteristics of the system, not just policy.
- **Co-authored outputs** — the transferable sovereign-AI framework, shaped with ACMI and shareable across the sector.
- **RMIT student projects (WIL)** — extending the existing internship pipeline into ARCHAI-based briefs.
- **Funding & futures** — grants, and where appropriate future deployment or licensing pathways for the toolkit.

05 What we'd explore together

Questions for the room — this is a collaboration to shape, not a product to install:

? Which collection subset makes the best low-risk pilot — born-digital, a permanent-gallery slice, a specific display?

? Which CMS / DAMS should ARCHAI read from, and what does the open collection API expose?

? What rights and licensing constraints apply to the records and media in scope?

- ? What cultural protocols and First Nations governance must be enforceable from day one?
- ? How does ACMI want to govern the new layer of visitor-conversation data?
- ? Is the appetite a light informal pilot, or a formal RMIT research partnership / ARC Linkage bid?
- ? Could ACMI host visitor studies to evaluate the living-archive model?

---

06 Why ACMI, why now

Rob knows ACMI from the inside — Visitor Experience, Public Programs, Exhibitions, Cinema Technician and AV roles, and the Nodel integration and permanent-gallery update — alongside Museums Victoria, Grande Experiences and the National Communication Museum, where he was most recently Head of Technology. ACMI's born-digital, screen-culture and time-based collections are exactly where sovereign preservation, replay and visitor-facing interpretation become most useful. And the moment is real: every cloud provider now has a pitch for museums. Sovereignty is not just an ethical preference — it is a preservation requirement.

Rob Graham · Creative technologist · Founder, Fine Art Media (FAMTEC) · PhD Researcher, RMIT University

rob@fineartmedia.tech · +61 409 336 251 · fineartmedia.tech/archai

Try it (best on a phone): fineartmedia.tech/acmi · the ACMI Object Wall concept

A concept demo and independent proposal — not affiliated with or endorsed by ACMI.